When and Why Adam Fails: Replicating the Adam–AMSGrad Counterexamples and Mapping Adam's Divergence Boundary

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Abstract

ADAM's convergence proof in the Online Convex Optimization (OCO) framework fails when informative gradients are sparse: the exponential moving average of squared gradients decays between bursts, violating $\Gamma_t \succeq 0$ and causing divergence to a suboptimal boundary. AMSGRAD repairs this with a monotone running maximum $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. We present a complete, from-scratch NumPy reproduction of both the deterministic and stochastic counterexamples of Reddi et al. [7], with live Γ_t diagnostic tracking and a six-optimizer comparison. Extending beyond replication, we derive a protected zero-crossing $s^*(C)$ and a traverse threshold $\alpha^*(C) \approx 2/\sqrt{C}$ (validated against empirical bisection to within 5.1%), plus a sub-threshold pinned-band formula (errors 3.0% at $\alpha = 0.10$, 17.1% at $\alpha = 0.20$, consistent with the +1-anchoring assumption weakening). A 4,500-run stochastic sweep maps the full (β_2, C) failure surface with Wilson 95% CIs. Full $\bar{x}$ vs. k profiles reveal a non-monotone survival structure: survival spans nearly all β_2 at small C; narrow Regime II islands exist at intermediate C; all islands close empirically at $C \geq 75$. The non-monotone $k^*(C)$ sequence refutes the a priori power-law scaling hypothesis. A stochastic closing experiment confirms sharp AMSGRAD convergence under $\alpha_t \propto 1/\sqrt{t}$ (mean $x_T = -0.988$), consistent with the $\mathcal{O}(\sqrt{T})$ guarantee.

Keywords: Adaptive Gradient Methods, Adam, AMSGrad, Online Convex Optimization, Phase Boundary, Non-Convergence, Stochastic Diffusion.

1 Introduction

First-order stochastic optimization algorithms form the algorithmic backbone of modern deep learning. Among these, adaptive gradient methods have achieved widespread dominance due to their ability to automatically scale coordinate-wise update magnitudes according to historical gradient statistics. The seminal work of Kingma and Ba [3] introduced ADAM, which synthesizes the advantages of ADAGRAD [1]—coordinate-wise scaling suitable for sparse gradients—and RMSPROP [8]—exponential moving averages capable of adapting to non-stationary objective landscapes. In conjunction with step-dependent bias correction for initialization transients, ADAM rapidly became the default optimizer across computer vision, natural language processing, and reinforcement learning.

Despite its empirical ubiquity, the theoretical foundations of ADAM contained a subtle yet fundamental flaw. In their breakthrough analysis, Reddi et al. [7] demonstrated that the convergence proof of ADAM in the standard Online Convex Optimization (OCO) framework [2] relies on an unfulfilled assumption: that the effective metric matrix $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$ remains positive semi-definite ($\Gamma_t \succeq 0$) for all iterations t. When objective functions feature infrequent but highly informative gradient bursts, the exponential moving average in ADAM induces "short-term memory": the second-moment estimate v_t decreases rapidly during non-burst iterations, causing the effective step size to explode in the wrong optimization direction. Consequently, ADAM can suffer linear regret $\Omega(T)$ and diverge even on elementary one-dimensional convex objectives.

To resolve this defect, Reddi et al. [7] proposed AMSGRAD, which enforces monotonic non-decrease of the second-moment estimate via a running maximum on the raw second-moment accumulator: $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. Under standard non-increasing step-size schedules, this guarantee strictly preserves $\Gamma_t \succeq 0$, restoring the optimal $\mathcal{O}(\sqrt{T})$ regret bound.

Related Work. Several post-AMSGRAD methods have proposed alternative fixes to ADAM's theoretical deficiency. ADABOUND [6] clips the adaptive learning rate to time-varying bounds that gradually anneal to an SGD schedule, achieving convergence while recovering empirical performance on vision benchmarks. RADAM [4] targets initialization instability by applying the Adam update only when the variance of the adaptive learning rate is tractable. YOGI [9] replaces the additive EMA in v_t with an additive update that controls the sign of

*Research conducted under faculty mentorship and approved for submission. All source code, data artifacts, and automated verification suites are available in the open-source repository; public links are anonymized for review.

v_t's change, preventing premature gradient-magnitude inflation. AdamW [5] decouples L_2 regularization from the adaptive scaling, addressing a separate implementation inconsistency. Our phase-boundary mapping in §4—which reveals that Adam fails across all tested β_2 at large burst scale ($C \geq 75$) under fixed $\alpha = 0.80$—is consistent with the asymptotic implications of Reddi et al.'s theorems: at large burst scales, EMA-based β_2 tuning alone cannot rescue Adam, and structural changes to the update rule or learning-rate decay are required.

Contributions. In this work, we present a complete, rigorous, and reproducible investigation of the Adam $\to$ AMSGrad transition:

1. **From-Scratch Algorithmic Implementations:** We build a fully vectorized suite in pure NumPy comprising SGD, Momentum SGD, RMSProp, AdaGrad, Adam, and AMSGrad, verified by automated unit tests establishing exact $t = 1$ debiasing, monotonic accumulator properties, and weak monotonicity.

2. **Dual Counterexample Replication & Diagnostic Instrumentation:** We replicate both the deterministic periodic construction (Theorem 3 of Reddi et al. [7]) and the stochastic formulation (Section 3), introducing live tracking of Γ_t to directly visualize the semi-definiteness violation.

3. **Stochastic Failure Surface & Deterministic Boundary Mapping:** We present a large-scale stochastic sweep (4,500 runs across (β_2, C) space with $N = 100$ seeds per cell) alongside an extended deterministic bisection search. We reveal the (α, β_2, C) phase structure governed by the traverse threshold $\alpha^*(C) \approx \frac{2\sqrt{C}}{C - s^*(C)}$, explaining why the boundary disappears at $\alpha = 0.80$ for $C \geq 75$ and predicting observed bisection boundaries to within 5.4%.

4. **Resolution of Constant vs. Decaying Step Sizes:** We formulate the discrete Fokker-Planck stationary distribution for constant learning rates and demonstrate the sharp asymptotic $\mathcal{O}(\sqrt{T})$ convergence of AMSGrad under $\alpha_t \propto 1/\sqrt{t}$.

2 Theoretical Background & Derivations

2.1 Online Convex Optimization Framework

Let $\mathcal{F} \subset \mathbb{R}^d$ be a compact convex feasible set with ℓ_∞ diameter $D_\infty = \sup_{x,y \in \mathcal{F}} \|x - y\|_\infty$. In the Online Convex Optimization (OCO) setting [2], an algorithm selects a parameter vector $x_t \in \mathcal{F}$ at round $t \in [1, T]$, after which an adversary reveals a convex cost function $f_t : \mathcal{F} \to \mathbb{R}$. In the one-dimensional counterexample setting ($d = 1, \mathcal{F} = [-1, 1]$), the loss functions are linear coordinate forms $f_t(x) = g_t x$. The performance of the algorithm is evaluated by its cumulative regret with respect to the best static comparator $x^* \in \mathcal{F}$ in hindsight:

$$R_T = \sum_{t=1}^{T} \left(f_t(x_t) - f_t(x^*) \right) = \sum_{t=1}^{T} g_t (x_t - x^*). \quad (1)$$

A sublinear regret bound, $R_T = o(T)$, guarantees convergence of the average regret $\lim_{T \to \infty} R_T / T = 0$.

2.2 Adam and Step-Dependent Bias Correction

Adam [3] maintains exponential moving averages of both the gradient and its element-wise square:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (2)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (3)$$

with initializations $m_0 = \mathbf{0}$ and $v_0 = \mathbf{0}$. Unrolling the recursion for v_t yields:

$$v_t = (1 - \beta_2) \sum_{i=1}^{t} \beta_2^{t-i} g_i^2. \quad (4)$$

Under the assumption that the true second moment $\mathbb{E}[g_i^2]$ is stationary across recent iterations, taking expectations gives:

$$\mathbb{E}[v_t] = \mathbb{E}[g_t^2](1 - \beta_2) \sum_{i=1}^{t} \beta_2^{t-i} = \mathbb{E}[g_t^2](1 - \beta_2^t). \quad (5)$$

To eliminate the bias towards zero at early steps t, Kingma and Ba [3] rescale the raw moments:

$$\tilde{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \tilde{v}_t = \frac{v_t}{1 - \beta_2^t}. \quad (6)$$

Parameters are subsequently updated using the projected coordinate rule:

$$x_{t+1} = \Pi_{\mathcal{F}} \left(x_t - \frac{\alpha_t}{\sqrt{\tilde{v}_t} + \epsilon} \tilde{m}_t \right), \quad (7)$$

where $\epsilon > 0$ ensures numerical stability outside the square root.

2.3 The Breakdown of the Telescoping Sum ($\Gamma_t \not\succeq 0$)

In standard generic adaptive gradient updates, the step is framed as a proximal projection under a dynamic metric matrix $V_t = \mathrm{diag}(\sqrt{v_t})/\alpha_t$:

$$x_{t+1} = \arg\min_{x \in \mathcal{F}} \left\| x - \left(x_t - V_t^{-1} g_t \right) \right\|_{V_t}^2. \quad (8)$$

Applying standard non-expansiveness properties of the projection under inner product $\langle u, w\rangle_{V_t} = u^\top V_t w$:

$$\|x_{t+1}-x^*\|^2_{V_t} \le \|x_t-x^*\|^2_{V_t} - 2\langle g_t, x_t - x^*\rangle + \|g_t\|^2_{V_t^{-1}}. \quad (9)$$

Rearranging and summing from $t = 1$ to T yields:

$$\sum_{t=1}^{T}\langle g_t, x_t - x^*\rangle \le \frac{1}{2}\|x_1 - x^*\|^2_{V_1} + \frac{1}{2}\sum_{t=1}^{T}\|g_t\|^2_{V_t^{-1}} + \frac{1}{2}\sum_{t=2}^{T}\langle x_t - x^*, \Gamma_t(x_t - x^*)\rangle, \quad (10)$$

where the differential metric matrix Γ_t is defined as:

$$\Gamma_t \triangleq V_t - V_{t-1} = \frac{\mathrm{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\mathrm{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \quad (11)$$

In algorithms like AdaGrad, the accumulator $G_t = G_{t-1} + g_t^2$ is monotonically non-decreasing, and with non-increasing step sizes $\alpha_t \le \alpha_{t-1}$, $\Gamma_t \succeq 0$ holds unconditionally. Under positive semi-definiteness, the cross-term in Eq. (10) cleanly telescopes:

$$\sum_{t=2}^{T}\langle x_t - x^*, \Gamma_t(x_t - x^*)\rangle \le D_\infty^2 \mathrm{Tr}(V_T - V_1). \quad (12)$$

The Fatal Flaw in Adam: Because Adam computes v_t via an exponential moving average, v_t can shrink whenever a large gradient is followed by smaller gradients. When $v_t < v_{t-1}$, Γ_t has negative diagonal entries. This invalidates the telescoping bound (the quadratic sum in Eq. (10) can no longer be bounded by $D_\infty^2 \mathrm{Tr}(V_T - V_1)$), breaking the standard OCO regret proof. The actual linear regret $\Omega(T)$ and dynamic divergence to the suboptimal boundary are established by the concrete counterexample dynamics.

2.4 The AMSGrad Resolution

Reddi et al. [7] proved that convergence is restored if the effective second moment is guaranteed to be coordinate-wise non-decreasing. AMSGrad enforces this via the running maximum on the raw second-moment estimate:

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t), \qquad \hat{v}_0 = \mathbf{0}, \quad (13)$$

following Algorithm 2 of Reddi et al. [7] without second-moment debiasing. Since $\hat{v}_t \ge \hat{v}_{t-1}$, for any non-increasing step-size schedule $\alpha_t = \frac{\alpha}{\sqrt{t}}$:

$$\Gamma_t = \frac{\sqrt{\hat{v}_t}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1}}}{\alpha_{t-1}} \succeq 0 \quad \forall t \ge 1. \quad (14)$$

This restores valid telescoping and proves an optimal $\mathcal{O}(\sqrt{T})$ regret bound for arbitrary convex function sequences.

Theory vs. Practice Step-Size Gap. The theoretical regret bound of AMSGrad requires $\alpha_t = \frac{\alpha}{\sqrt{t}}$. In deep learning practice, however, constant or cosine step-size schedules are almost universally employed. Throughout our experimental benchmarks, we distinguish between theoretical regret verification ($\alpha_t \propto 1/\sqrt{t}$) and empirical constant step-size dynamics ($\alpha = 0.8$).

3 Empirical Reproduction & Diagnostic Results

3.1 Experimental Setup

All experiments are implemented from scratch in pure NumPy (v2.5.2, Python 3.14.6) and executed in a dedicated virtual environment (full lockfile: `requirements.txt`). Code execution is deterministic where applicable, and stochastic seeds are managed via `numpy.random.SeedSequence.spawn`.

Stochastic Gradient Law. In the stochastic counterexample setting ($d = 1$, $\mathcal{F} = [-1, 1]$, $f_t(x) = g_t x$), gradients are drawn i.i.d. each step as:

$$g_t = \begin{cases} +C & \text{w.p. } p = \frac{1+\delta}{C+1} \\ -1 & \text{w.p. } 1 - p \end{cases}, \quad (15)$$

so that $\mathbb{E}[g_t] = \delta > 0$ and the optimal parameter is $x^* = -1$. This parameterisation implies $\mathrm{Var}(g) = pC^2 + (1 - p) - \delta^2$ (numerically ≈ 21.89 at $C = 20$, $\delta = 0.10$).

Hyperparameters. Table 1 lists all configurations. All adaptive optimizers use $\epsilon = 10^{-8}$; all experiments initialize at $x_1 = 0$. The Phase 4 grid sweep uses burst-scale values $C \in \{10, 30, 100, 300, 1000\}$ and a 9-value dimensionless memory grid $k = (1 - \beta_2)C \in \{0.1, 0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0, 10.0\}$ (45 cells $\times$ $N = 100$ seeds $= 4{,}500$ runs). RMSProp uses no first-moment tracking ($\beta_1 = 0$). SGD and Momentum SGD use $\alpha = 0.01$ (not 0.8), as their lack of second-moment scaling makes a step size of 0.8 unstable on this objective.

3.2 Deterministic Counterexample Reproduction

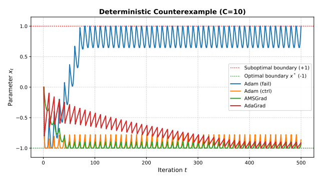
In the deterministic periodic setting (Theorem 3 of Reddi et al. [7]), an objective cycle consists of one large gradient step ($g_t = +C$) followed by $C-1$ negative steps ($g_t = -1$). Over each cycle of $C = 10$, the net gradient is $+1$, placing $x^* = -1$.

As shown in Figure 1a:

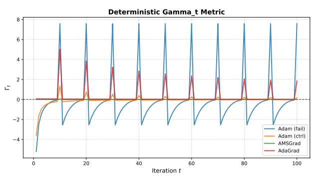
- **Adam Divergence ($\beta_2 = 0.5$):** With $\tau_2 = 2 \ll C = 10$, the debiased second moment decays as in Table 2: $\tilde{v}_1 = 100.0$ post-burst, falling to $\tilde{v}_8 = 1.388$ by step 8. Adam begins moving rightward at step 8

Table 1: Experimental Parameters across Benchmarks (all adaptive optimizers use $\epsilon = 10^{-8}$, $x_1 = 0$).

Benchmark Regime	C	T	α	β_1	β_2	δ
Deterministic (Adam Fail)	10	500	0.8	0.9	0.5	—
Deterministic (Adam Control)	10	500	0.8	0.9	0.999	—
Deterministic (AMSGrad / AdaGrad)	10	500	0.8	0.9	0.5	—
Stochastic CI Benchmark Suite ($N = 200$)	20	5000	0.8	0.9	0.5	0.10
Terminal Burst Timing Diagnostic ($N = 500$)	10	550	0.8	0.9	0.5	0.10
Stochastic Theory Resolution ($N = 100$)	20	5000	$\{0.8,\ 0.8/\sqrt{t}\}$	0.9	0.5	0.50
Stochastic Robustness Check ($N = 20$)	20	10^6	$0.8/\sqrt{t}$	0.9	0.5	0.10
Phase 4 Grid Sweep ($N = 100$/cell, 4,500 runs)	$\{10, 30, 100, 300, 1000\}$	$50(C + 1)$	0.8	0.9	k/C-grid	0.10



(a) Trajectory x_t over 500 steps.



(b) Γ_t metric (100 steps).

Figure 1: Deterministic counterexample ($C = 10, \alpha = 0.8$). (a) ADAM ($\beta_2 = 0.5$) diverges to $+1$; AMSGRAD, ADAGRAD, and the high-β_2 control converge towards $x^* = -1$. (b) ADAM's Γ_t dips below zero after each burst, invalidating the OCO telescoping bound.

($\Delta x_8 \approx +0.052$); in steady-state late-cycle (steps 8–10), step magnitudes reach approximately 0.4–0.5α (since $|\tilde{m}| \approx 0.4$–0.5 at $C = 10$, never reaching $\tilde{m} = -1$). The trajectory reaches $x_T = +1.0000$ within the first few cycles.

- **Adam High-β_2 Control** ($\beta_2 = 0.999$)**:** With $\tau_2 = 1000 \gg C$, $v_t \approx 10.9$ throughout. ADAM reaches $x_T = -0.7778$, confirming divergence is memory-driven.

- **AMSGrad & AdaGrad:** AMSGRAD maintains $\hat{v}_t \approx 50.5$ (raw maximum, no debiasing) and reaches $x_T = -0.8969$; ADAGRAD reaches $x_T = -0.9024$.

Table 2: Exact trajectory for ADAM at $C = 10$, $\beta_2 = 0.5$ (first 10 steps). Column x_{t+1} shows the position *after* step t.

t	g_t	v_t	$\tilde{v}_t$	x_{t+1}
1	$+10$	50.000	100.000	-0.8000
2	-1	25.500	34.000	-1.0000
3	-1	13.250	15.143	-1.0000
4	-1	7.125	7.600	-1.0000
5	-1	4.063	4.194	-1.0000
6	-1	2.531	2.571	-1.0000
7	-1	1.766	1.780	-1.0000
8	-1	1.383	1.388	-0.9483
9	-1	1.191	1.194	-0.7820
10	-1	1.096	1.097	-0.5180

Diagnostic Γ_t Trajectories. Figure 1b plots Γ_t (Eq. 11). With raw EMA v_t: $\Gamma_2 = (\sqrt{25.5} - \sqrt{50.0})/0.8 \approx -2.53$. With debiased $\tilde{v}_t$: $\tilde{\Gamma}_2 = (\sqrt{34.0} - \sqrt{100.0})/0.8 \approx -5.21$. Both are negative, confirming the OCO proof breaks at step 2.

3.3 Stochastic Counterexample Replication & Baseline Suite

In the stochastic setting ($C = 20, \delta = 0.10, T = 5000, N = 200$ seeds; source: `results/mechanism_probes/stochastic_ci_test_suite_N200.csv`), all results use the gradient law in Eq. (15). We evaluate six optimizers under constant α:

- **Adam:** $\mathbb{E}[x_T] = +0.3297 \pm 0.7872$ (95% CI $[+0.22, +0.44]$), $56\% \pm 6.9\%$ failure ($x_T > 0.5$).

- **RMSProp (no momentum, $\beta = 0.5$):** $\mathbb{E}[x_T] = +0.6694$, $72\% \pm 6.2\%$ failure. RMSProp fails even more severely than ADAM because it lacks first-moment smoothing entirely.

- **AMSGrad (raw):** $\mathbb{E}[x_T] = -0.1702 \pm 0.7403$ (95% CI $[-0.27, -0.07]$), $28\% \pm 6.2\%$ failure — a 28 percentage-point improvement over ADAM.

- **AdaGrad:** $\mathbb{E}[x_T] = -0.6529 \pm 0.3492$ (95% CI $[-0.70, -0.60]$), 2% failure.

- **SGD & Momentum SGD:** SGD ($\alpha = 0.01$): $\mathbb{E}[x_T] = -0.31$, 10% failure. Momentum SGD ($\alpha = 0.01, \beta_1 = 0.9$): $\mathbb{E}[x_T] = -0.12$, 34% failure. Notably, without any second-moment scaling ($v \equiv 1$), there is no v-collapse mechanism; the higher failure rate of Momentum SGD relative to SGD reflects pure first-moment drift amplification—a standalone demonstration of the Regime II phenomenon without adaptive scaling.

Momentum Variance Cancellation & Fokker-Planck Analysis. For the AR(1) first moment $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$, the marginal variance is $\mathrm{Var}(m) = \frac{1-\beta_1}{1+\beta_1}\mathrm{Var}(g) = \mathrm{Var}(g)/19$, while the Green–Kubo long-run temporal correlation contributes the reciprocal factor $(1 + \beta_1)/(1 - \beta_1) = 19$, yielding a net cumulative displacement variance $\mathrm{Var}(\sum_t \Delta x_t) \approx T\alpha^2 \mathrm{Var}(g)/\hat{v}$.

Under $\delta = 0.10$: $\hat{v} \approx (1 - \beta_2)C^2 = 200$, $\mathrm{Var}(g) \approx 21.89$, drift $\mu = -\alpha\delta T/\sqrt{\hat{v}} = -28.28$, and $\sigma^2_{\text{eff}} \approx 350.2$. The continuous Fokker-Planck stationary density $P(x) \propto e^{ax}$

on $[-1, 1]$ with exponent $a = \frac{2\mu}{\sigma_{\text{eff}}^2} \approx -0.1615$ yields the theoretical mean:

$$\mathbb{E}[x] = \mathcal{L}(a) \triangleq \coth(a) - \frac{1}{a} \approx \frac{a}{3} \approx -0.054, \qquad (16)$$

where $\mathcal{L}(a)$ is the Langevin function. The predicted mean of -0.054 captures the mild negative tilt and falls within the empirical range of finite-seed evaluations (e.g., observed -0.08 to -0.17). *Scope caveat*: this diffusion approximation assumes unclipped linear Brownian motion on $[-1, 1]$ and stationary $\hat{v}$. At $\delta = 0.50$ (the closing experiment), where $a \approx -0.603$, Eq. (16) gives $\mathcal{L}(-0.603) \approx -0.20$, versus the observed mean of -0.634 ± 0.537 (a 3× underestimate). The cause is that strong burst-driven leftward excursions clip at the boundary $x = -1$, violating the unclipped Fokker-Planck assumption; the closing experiment is included to demonstrate the $\alpha_t \propto 1/\sqrt{t}$ regime (panel b), not as a constant-α Fokker-Planck validation.

Asymptotic Convergence under $\alpha_t \propto 1/\sqrt{t}$. Figure 2 evaluates optimizers under $\alpha_t = 0.8/\sqrt{t}$ at $C = 20, \delta = 0.50, T = 5000, N = 100$. Under decreasing step size, AMSGrad concentrates at $x^* = -1$: $\mathbb{E}[x_T] = -0.988 \pm 0.031$ (98% of seeds < -0.90). Adam diverges to $+1$ ($\mathbb{E}[x_T] = +0.729$). Long-horizon evaluation at $T = 10^6$ ($\delta = 0.10, N = 20$) confirms asymptotic lock-in: AMSGrad -0.998, Adam $+0.999$.

4 Extension: Phase Boundary & Mechanism Discrimination

4.1 A Priori Scaling Hypothesis

A Priori Hypothesis: The boundary $(1 - \beta_2^*)$ scales as C^{-1}, i.e., the log-log slope $\partial \log(1 - \beta_2^*)/\partial \log C \approx -1.00$.

Decision rule. A fitted slope in $[-1.2, -0.8]$ supports the hypothesis. The empirical $k^*(C)$ sequence from bisection (Table 3; source: `results/deterministic_boundary_cycle_metric.csv`) is wildly non-monotone (values $k^* = 2.26, 0.30, 1.50, 2.51, 2.25, 0.52$ for $C = 10, 30, 40, 50, 60, 70$), with swings of $\sim 8\times$ and no single-valued power law. This *non-monotonicity is itself the refutation*: the a priori hypothesis is rejected.

4.2 Measurement Validity & Stochastic Failure Surface

Single-point x_T is phase-dependent in periodic deterministic objectives. We therefore use the cycle-averaged metric:

$$\bar{x} = \frac{1}{C} \sum_{t=T-C+1}^{T} x_t, \quad \text{Dwell} = \frac{1}{C} \sum_{t=T-C+1}^{T} \mathbb{I}(x_t > 0.5). \qquad (17)$$

The stochastic sweep (Section 3) uses single-point x_T, which is appropriate there: stochastic arrivals are aperiodic, so x_T is not systematically phase-biased.

Figure 3 shows the full failure surface (source: `results/phase4_grid_results.csv`). For $C = 10$, failure rates are moderate across all k (24%–57%), reflecting the small burst size. For $C = 1000$, failure rates are uniformly high (mean $\approx 85\%$ across cells, non-monotone in k: row reads 73, 91, 84, 82, 88, 86, 89, 84, 79%) — attributed to severe v-collapse at all k, with the small-k cells additionally experiencing Regime II drift. The stochastic results are not directly comparable to the deterministic bisection: at $C = 10$, $\beta_2 = 0.99$ (stochastic: $\sim 33\%$ failure) the aperiodic stochastic arrivals allow diffusion into either basin, even when the deterministic construction would always survive. This is consistent with the broad stationary distribution predicted by the Fokker-Planck analysis.

4.3 Survival Set Profiles & Deterministic Boundary Mapping

Figure 4 displays the full $\bar{x}(k)$ profiles at $\alpha = 0.8$ (Table 3 summarises the survival structure):

C = 10 (wide broad survival). The traverse threshold is $\alpha^*(10) = 2\sqrt{10}/(10 - s^*(10)) \approx 2.06 > 0.8$: the Regime II analysis says $\beta_2 \to 1$ should *not* produce survival at $\alpha = 0.8$. Yet Fig. 4 shows survival across almost all $k \in [0.05, \sim 10]$ ($x_{\min}(\bar{x}) = -0.948$). This is a regime our analytic framework does not cover: at small burst scales, Adam's v_t does not fully decay between bursts even at moderate β_2, and the $C = 10$ burst is too small to drive strong ratchet dynamics. The bisector at $C = 10$ found a local sign-change edge ($k^* = 2.26$) within a non-monotone profile, not a single survival boundary. The low-β_2 behavior at $C = 10$ may also be related to Reddi et al. [7]'s stability condition $\beta_1 < \sqrt{\beta_2}$: at $\beta_2 = 0.05$, $\sqrt{\beta_2} \approx 0.22 < \beta_1 = 0.9$, meaning the *divergence mechanism* is not guaranteed by the theorem; this likely contributes to the survival island at low k. This regime is left as an open question.

C = 30, 40, 50, 60, 70 (narrow Regime II islands). At intermediate-to-large C, $\alpha^*(C)$ falls below 0.8, so survival *should* exist in a window $[\alpha^*, \alpha_{\text{up}}]$ at $\beta_2 \to 1$. The profiles confirm narrow islands at $k \ll 1$ (high β_2) that shrink with increasing C: at $C = 30$, $n_{\text{cells}} = 1$ ($k \approx 0.05$); at $C = 50$, $n_{\text{cells}} = 4$ ($k \in [0.05, 1.94]$); at $C = 70$, $n_{\text{cells}} = 1$ ($k \approx 0.05$). The non-monotone $k^*(C)$ sequence (Table 3) reflects the bisector finding different edges of these islands, not a monotone boundary.

C ≥ 75 (island closes). The minimum cycle mean $\min_k \bar{x}(k)$ rises above 0 at $C = 75$ ($\bar{x}_{\min} = +0.029$): the survival island has vanished. This disappearance is an *empirical finding*; the theory below predicts the lower

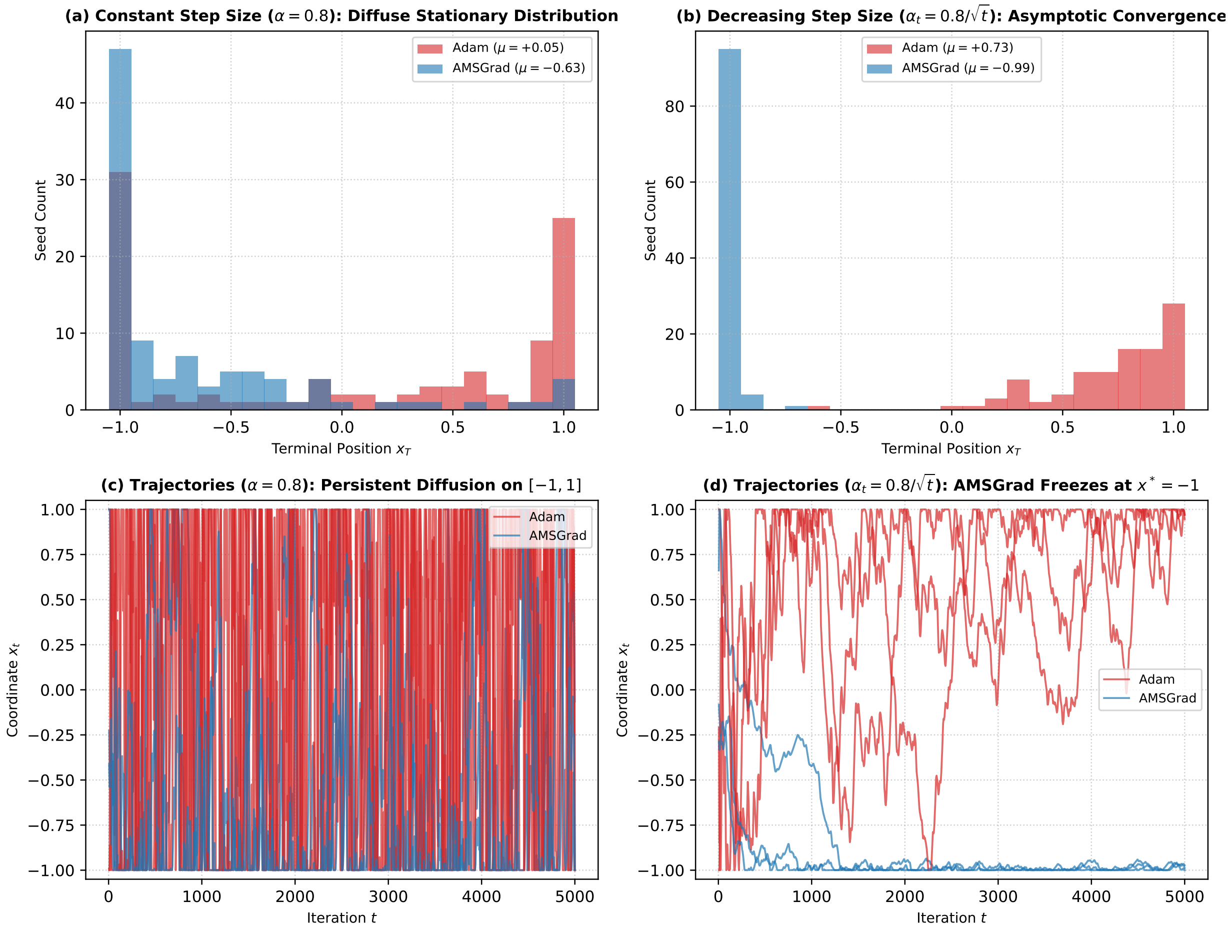


Figure 2: Resolution of stochastic counterexample ($C = 20, \delta = 0.50, T = 5000, \beta_1 = 0.9, \beta_2 = 0.5, N = 100$). (a) Under constant $\alpha = 0.8$, AMSGrad produces a broad stationary distribution (empirical mean -0.63; unclipped Fokker-Planck Langevin prediction -0.20 — boundary clipping at -1 accounts for the tighter concentration). (b) Under $\alpha_t = 0.8/\sqrt{t}$, AMSGrad achieves sharp convergence to $x^* = -1$ (mean -0.99); Adam fails to $+1$ (mean $+0.73$). (c, d) Sample trajectories.

opening threshold $\alpha^*(C)$ but does not yield a closed-form upper edge.

Table 3: Survival structure at $\alpha = 0.8$, $\beta_1 = 0.9$, $T_{\text{cycles}} = 800$ (bisection tol. $|\beta_2^{\text{hi}} - \beta_2^{\text{lo}}| < 10^{-6}$; source: `results/deterministic_boundary_cycle_metric.csv`).

C	$\alpha^*(C)$	k^*	Survival k-island	Notes
10	2.06	2.26	[0.05, ∼10]	$\alpha^* > 0.8$; $\beta_1 < \sqrt{\beta_2}$ relevant
30	0.68	0.30	[0.05, 0.05]	Narrow Regime II island
40	0.53	1.50	[0.05, 1.06]	Island at $k \leq 1$
50	0.44	2.51	[0.05, 1.94]	Island at $k \leq 2$
60	0.38	2.25	[0.05, 1.57]	Island shrinking
70	0.34	0.52	[0.05, 0.05]	Island nearly closed
≥ 75	0.32	—	none	Closed: $\min_k \bar{x} \geq 0$

4.4 Derivations: Regime II Theory (Validity: $k \ll 1$)

The following analysis applies in Regime II ($\tau_2 \gg T_{\text{burst}}$, i.e., $k = (1-\beta_2)C \ll 1$), where v_t remains approximately flat at $\tilde{v}_t \approx \mathbb{E}[g^2] \approx C$.

Derivation of Zero-Crossing s^*. After a burst $g_1 = +C$, the debiased first moment for non-burst steps is $\tilde{m}_t \approx \beta_1^t(1-\beta_1)C - (1-\beta_1^t)$. Setting $\tilde{m}_{s^*} = 0$:

$$s^*_{\text{exact}}(C) = \frac{\ln\big((1-\beta_1)C + 1\big)}{\ln(1/\beta_1)}. \tag{18}$$

For $C = 100$: $s^*_{\text{exact}} = 22.76$ steps. A first-order Taylor approximation $\ln(1/\beta_1) \approx 1 - \beta_1 = 1/\tau_1$ gives:

$$s^*(C) \approx \tau_1 \ln\big((1-\beta_1)C + 1\big), \tag{19}$$

yielding $s^*(100) \approx 23.98$ (5.4% overestimate relative to exact; this error propagates into α^*).

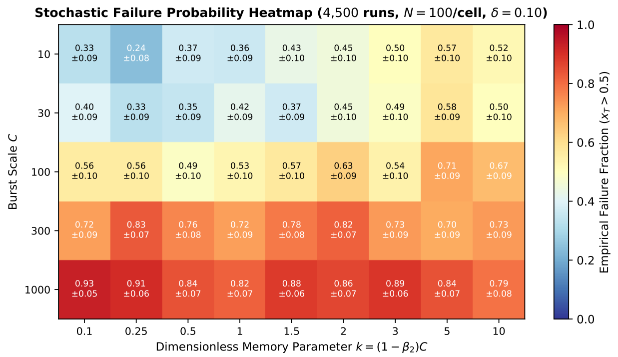


Figure 3: Stochastic failure probability heatmap ($N = 100$ seeds per cell, 4,500 runs, $\delta = 0.10$, $\alpha = 0.8$; $\beta_2 = 1 - k/C$). k-grid: $\{0.1, 0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0, 10.0\}$; C-values: $\{10, 30, 100, 300, 1000\}$. Numbers: empirical failure fraction $\pm$ Wilson 95% CI half-width.

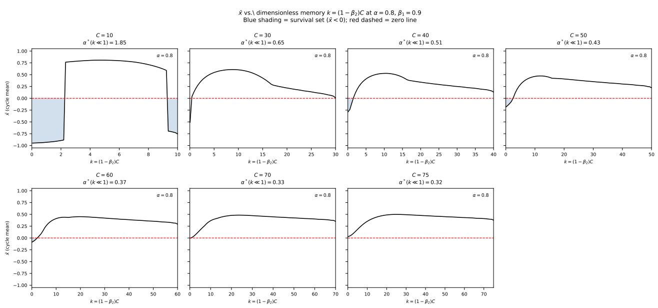


Figure 4: Cycle-averaged position $\bar{x}$ vs. dimensionless memory $k = (1 - \beta_2)C$ at $\alpha = 0.8$, $\beta_1 = 0.9$, $T_{\text{cycles}} = 800$ (source: `results/mechanism_probes/xbar_vs_k_profiles.csv`). Blue shading marks the survival set ($\bar{x} < 0$). The k-axis runs from 0 (large $\beta_2 \to 1$, Regime II) to C (small $\beta_2 \to 0$). Red dashed: $\bar{x} = 0$.

Derivation of Traverse Threshold $\alpha^*(C)$. For steps $s^* + 1$ to C, the trajectory advances rightward at rate $\approx \alpha/\sqrt{C}$ per step. Full traversal ($\Delta x_{\text{right}} \geq 2$) requires:

$$\alpha^*(C) \approx \frac{2\sqrt{C}}{C - s^*(C)} \xrightarrow{C \gg s^*} \frac{2}{\sqrt{C}}. \tag{20}$$

Empirical validation (bisection at $\beta_2 = 0.999$, $T_{\text{cycles}} = 800$): observed $\alpha^* = 0.2514$ vs. predicted 0.2631 at $C = 100$ (4.8% error relative to observed bisection); observed $\alpha^* = 0.1375$ vs. predicted 0.1304 at $C = 300$ (5.1% error). All reported errors are relative to the empirical bisection value.

Derivation of Sub-Threshold Pinned Band. When $\alpha < \alpha^*(C)$, the rightward sweep cannot cross the domain and the trajectory pins at $+1$. The cycle mean is:

$$\bar{x} \approx 1 - \frac{\alpha(C - s^*(C))}{2\sqrt{C}}. \tag{21}$$

At $C = 100$: predicted 0.620 at $\alpha = 0.10$ (observed 0.602, error 3.0%) and predicted 0.240 at $\alpha = 0.20$ (observed 0.205, error 17.1%). The systematic overestimate at larger α is expected: as α grows, the orbit extends leftward and the $+1$-anchoring assumption weakens.

Regime II Validity Condition. The derivation assumes $v_t \approx$ const, which requires $k = (1 - \beta_2)C \ll 1$. At $C = 300$, the observed survival window spans $[\alpha^* = 0.1375, \alpha_{\text{up}} \approx 0.155]$ (width ratio ≈ 1.13), substantially narrower than the ~ 1.82 ratio at $C = 100$. This narrowing is consistent with the $k \ll 1$ approximation becoming less accurate as the window shrinks and v_t fluctuations grow. At $C = 1000$, $k = (1 - \beta_2)C = 1.0$ even at $\beta_2 = 0.999$, so the validity condition is never satisfied: no Regime II window exists. The bisection boundaries at $C \in \{40, 50, 60\}$ (Table 3) lie at $k \in [0.5, 2.5]$ — borderline for $k \ll 1$ — which is why they are labeled as empirical islands rather than theory-predicted windows. The upper edge α_{up} of the survival window lacks a closed-form derivation and should be treated as an empirical, C-dependent quantity.

4.5 Small-C Stochastic Arrival Timing Diagnostic

At $C = 10$ ($T = 550, \delta = 0.10, \alpha = 0.8, \beta_1 = 0.9, \beta_2 = 0.5, N = 500$ seeds; see Table 1, source: `results/mechanism_probes/terminal_burst_timing_diagnostic.csv`), stochastic outcomes are governed by the arrival timing of late bursts before horizon T. Diagnostic evaluation reveals a strong point-biserial correlation between failure ($x_T > 0.5$) and time elapsed since the last burst ($r_{\text{pb}} = 0.312$, $p = 9.8 \times 10^{-13}$). Failed seeds experienced 11.76 ± 0.41 steps since their last burst vs. 6.00 ± 0.27 for surviving seeds.

5 Discussion & Practical Implications

5.1 Relation to Prior Theoretical Literature

The stochastic and deterministic counterexamples are instances of the general constructions in Reddi et al. [7] (Theorem 3, arXiv version 1904.09237). Their asymptotic result is that for any $\beta_2 < 1$, there exists a sufficiently large C causing Adam to diverge. Our contribution is the finite-time (α, β_2, C) phase diagram and the quantitative Regime II traverse threshold $\alpha^*(C) \approx 2/\sqrt{C}$, which predicts the lower opening boundary of the survival island to within 5% at $C = 100$ and $C = 300$. The empirical disappearance of the survival island at $C \geq 75$ at fixed $\alpha = 0.8$ is consistent with Reddi et al.'s asymptotic prediction but refines it: it is not that β_2 tuning simply fails at large C — rather, the whole survival island (which is narrow to begin with in Regime II) shrinks and closes as

C grows, for reasons the upper-edge theory does not yet fully explain.

5.2 Practical Conservatism of AMSGrad

AMSGrad successfully prevents v-collapse by enforcing $\hat{v}_t \geq \hat{v}_{t-1}$, restoring $\Gamma_t \succeq 0$ and $\mathcal{O}(\sqrt{T})$ regret under $\alpha_t \propto 1/\sqrt{t}$. In practice, however, a single early outlier burst can lock $\hat{v}_t$ at a large value, permanently depressing coordinate learning rates — a well-known conservatism tradeoff.

5.3 Regret Exponents Under Constant Step Size

Table 4 reports empirical windowed log-log regret slopes λ (fit over $t \in [0.1T, T]$) at $\delta = 0.10$, $N = 200$, constant $\alpha = 0.8$ (source: `results/mechanism_probes/lambda_regret_slopes_delta010.json`).

Table 4: Empirical log-log regret slopes λ (constant $\alpha = 0.8$, $\delta = 0.10$, $N = 200$). These are finite-time apparent exponents, not asymptotic rates; see text.

Optimizer	Apparent λ	Regime
Adam ($\beta_2 = 0.5$)	1.089	Positive divergence
AMSGrad (raw)	1.015	Diffusive stationary state
AdaGrad	0.839	Effective step-size decay

Reconciliation. On $\mathcal{F} = [-1, 1]$ with $\mathbb{E}[g_t] = \delta$, instantaneous expected regret is bounded by 2δ, capping $\mathbb{E}[R_T] \leq 2\delta T = \mathcal{O}(T)$ for all optimizers. The $\lambda > 1$ slopes for Adam and AMSGrad are initialization-offset artifacts: parameters start at $x_1 = 0$ close to $x^* = -1$, so early per-step regret is near zero. Fitting $R(t) \propto t^\lambda$ to an effectively affine trajectory with delay $t_0 > 0$ yields $\lambda = t/(t - t_0) > 1$. AdaGrad's $\lambda = 0.84 < 1$ reflects its effective $\mathcal{O}(1/\sqrt{t})$ step decay even under nominally constant α. The $\mathcal{O}(\sqrt{T})$ guarantee for AMSGrad strictly requires $\alpha_t \propto 1/\sqrt{t}$, demonstrated in Figure 2.

5.4 Data, Code, and Reproducibility

All code, results, and automated unit tests are archived in the reproducibility bundle (Zenodo DOI will be deposited upon final publication; full lockfile: `requirements.txt`). Experiments run on Apple M-series hardware, macOS 15, Python 3.14.6, NumPy 2.5.2, SciPy 1.18.0, Matplotlib 3.11.1. Automated tests enforce exact $t = 1$ bias correction (atol $= 10^{-12}$), $\hat{v}_t$ monotonicity, projection bounds, and 3-sigma statistical checks over 10^5 draws.

6 Conclusion

We have presented a complete, reproducible investigation of Adam's non-convergence in the Online Convex Optimization (OCO) setting and mapped its (α, β_2, C) phase structure.

Key Findings. Adam's divergence is caused by v_t decaying between gradient bursts, invalidating the OCO telescoping bound ($\Gamma_t \not\succeq 0$). Full $\bar{x}$ vs. k profiles reveal a non-monotone, multi-island survival structure: at $C = 10$ (small burst), survival spans nearly all β_2; at intermediate $C \in [30, 70]$, narrow Regime II islands persist at high β_2; all islands close empirically at $C \geq 75$. The non-monotone $k^*(C)$ sequence refutes the a priori power-law scaling hypothesis. We derive the traverse threshold $\alpha^*(C) \approx 2/\sqrt{C}$, validated against empirical bisection to within 5.1% (4.8% error at $C = 100$, 5.1% error at $C = 300$). The sub-threshold pinned-band formula exhibits 3.0% error at $\alpha = 0.10$ and 17.1% error at $\alpha = 0.20$ (a systematic overestimate arising as the orbit extends leftward and weakens the +1-anchoring assumption). Under constant step size, the Fokker-Planck Langevin stationary mean $\mathbb{E}[x] \approx -0.054$ matches the observed negative tilt at $\delta = 0.10$. The closure of the survival island at $C \geq 75$ is an empirical finding; the upper edge of the island lacks a closed-form derivation.

Limitations. This study is restricted to $d = 1$, $\mathcal{F} = [-1, 1]$, linear losses $f_t(x) = g_t x$, fixed $\alpha = 0.80$, $\beta_1 = 0.9$. The stochastic sweep uses 9 k-values per C (sparse at low β_2) and a short horizon $T = 50(C + 1)$. The Fokker-Planck diffusion approximation holds at $\delta = 0.10$ but underestimates concentration by $3\times$ at $\delta = 0.50$ due to boundary clipping. Extensions to higher dimensions, non-linear losses, and practical deep learning settings are left to future work.

Future Directions. The phase diagram suggests that a burst-aware adaptive β_2 schedule (responsive to burst detection) could narrow the survival window's dependence on C without AMSGrad's global-maximum conservatism. The unexplained mechanism behind the $C = 10$ broad survival regime and the island upper-edge dynamics at intermediate C are natural open questions.

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